Maintenance Model -- IBMS (Integrated Business Management Suite)

Field	Detail
Project Name	Integrated Business Management Suite (IBMS) v2.0
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Sprint	Sprint 1
Date	April 14, 2026
Document Type	Software Maintenance Model

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1. Introduction

This document defines the maintenance model for the IBMS platform, covering all post-deployment activities required to keep the system operational, secure, and aligned with evolving business needs. IBMS is a multi-layered enterprise system spanning FastAPI backend, Frappe framework integration, dual databases (MongoDB + MariaDB), Redis caching, real-time WebSocket streaming, AI/ML services, and multi-cloud deployment (Docker, Kubernetes, AWS, GCP).

The maintenance model follows a **hybrid approach** combining proactive scheduled maintenance with reactive incident response, aligning with IEEE 14764 software maintenance standards.

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2. Maintenance Categories

2.1 Overview

Category	Description	Estimated Effort	Frequency

Category	Description	Estimated Effort	Frequency
Corrective	Bug fixes, defect resolution, incident patches	30%	As needed (reactive)
Adaptive	Environment changes, dependency updates, platform migrations	25%	Monthly / Quarterly
Perfective	Feature enhancements, performance optimization, UX improvements	30%	Per sprint (bi-weekly)
Preventive	Code refactoring, tech debt reduction, proactive hardening	15%	Monthly

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2.2 Corrective Maintenance

Corrective maintenance addresses defects discovered in production or through testing.

Activity	Scope	Trigger	SLA
Critical bug fix (SEV-1)	System-down, data corruption, security breach	User report, monitoring alert	4 hours to fix, 1 hour to deploy
Major bug fix (SEV-2)	Feature broken, data inconsistency, high error rate	User report, automated tests	24 hours to fix
Minor bug fix (SEV-3)	UI defect, non-critical logic error, edge case	User report, QA testing	Next sprint
Cosmetic fix (SEV-4)	Typos, alignment, minor UI issues	User feedback	Backlog

Corrective Maintenance Workflow:

```
Bug Report -> Triage (Severity) -> Assign -> Root Cause Analysis -> Fix -> Test -> Review -> Deploy -> Verify ->
  Close
```

IBMS-Specific Corrective Focus Areas:

Component	Common Issue Types	Diagnostic Tools
JWT Auth (security/jwt_auth.py)	Token expiry edge cases, secret rotation failures	/api/debug/auth-bench, audit logs
Dual DB Layer (database/)	Cross-database inconsistency, connection pool exhaustion	/api/health, reconciliation jobs
AI Services (services/)	False positive anomalies, stale model predictions	AI Alert tracking, confidence metrics
WebSocket (/ws/kpi)	Connection drops, auth race conditions	WS connection counter in /api/health
Background Jobs (jobs/)	Missed schedules, Frappe queue deadlocks	Frappe scheduler logs, /api/metrics
Rate Limiter	False rate-limit blocks, Redis fallback misses	Rate limit logs, /api/system/status

2.3 Adaptive Maintenance

Adaptive maintenance addresses changes in the operating environment, dependencies, and platform requirements.

Activity	Scope	Schedule	Owner
Python dependency updates	Update requirements.txt packages (FastAPI, Pydantic, Motor, etc.)	Monthly	Developer
Frappe framework upgrades	Compatibility testing with upstream Frappe releases	Quarterly	Developer
Database engine updates	MongoDB 7.x, MariaDB 10.x patch updates	Quarterly	DevOps
Redis version updates	Redis 7.x patch updates, feature adoption	Quarterly	DevOps

Activity	Scope	Schedule	Owner
Docker base image updates	python:3.11-slim security patches	Monthly	DevOps
Kubernetes version updates	K8s API changes, deprecation handling	Semi-annually	DevOps
Cloud provider changes	AWS/GCP service updates, API changes, pricing adjustments	As announced	DevOps
SSL/TLS certificate renewal	Nginx TLS certificates for production	Before expiry (auto-renew preferred)	DevOps
OS-level security patches	Container base OS CVE patches	Monthly	DevOps

Dependency Update Process:

```
Scan (dependabot/pip-audit) -> Evaluate Impact -> Update -> Test Suite -> Staging Deploy -> Production Deploy
```

2.4 Perfective Maintenance

Perfective maintenance implements new features and optimizations planned in the product roadmap.

Activity	Example	Sprint Allocation
New API endpoints	Additional ERP modules (purchasing, HR, payroll)	Per roadmap
AI model improvements	Enhanced anomaly detection, new lead scoring features	Per sprint
UI/UX enhancements	Dashboard redesign, mobile responsiveness, accessibility	Per sprint
Performance optimization	Query optimization, caching strategy improvements	Per sprint
New integrations	Additional webhook providers, SSO providers, payment gateways	Per roadmap
Reporting enhancements	New KPI metrics, custom report builder, export formats	Per sprint

Feature Delivery Workflow:

```
User Story -> Sprint Planning -> Design -> Implement -> Unit Test -> Integration Test -> Code Review -> Staging
  -> UAT -> Production
```

2.5 Preventive Maintenance

Preventive maintenance proactively addresses tech debt and structural weaknesses before they become incidents.

Activity	Scope	Schedule	Benefit
Code quality audits	Lint, type checking, complexity analysis	Monthly	Reduce bug density
Database index optimization	Analyze slow queries, add/remove indexes	Monthly	Improve response times
Log rotation and archival	Rotate audit_logs, kpi_snapshots, webhook_logs	Weekly (automated)	Prevent storage exhaustion
Security vulnerability scanning	pip-audit, safety, container image scanning	Per commit (CI)	Prevent known CVE exposure
Dead code removal	Remove unused endpoints, deprecated services	Quarterly	Reduce attack surface
Configuration drift detection	Compare running config vs source-of-truth	Monthly	Prevent misconfiguration
Load testing	Simulate peak traffic against staging	Quarterly	Validate capacity planning
Backup integrity verification	Restore test from MongoDB/MariaDB backups	Monthly	Ensure recoverability

3. Maintenance Infrastructure

3.1 Existing Automated Maintenance Systems

IBMS already includes several automated maintenance capabilities:

System	Component	Function	Schedule
KPI Refresh	jobs/nightly_kpi_refresh.py	Full KPI metric recalculation	Daily at 3:00 AM
KPI Rollup	jobs/kpi_rollup.py	Hourly metric aggregation	Every hour
Compliance Audit	jobs/compliance_check.py	Automated compliance checks + recommendations	Daily at 3:30 AM
Workflow Optimization	jobs/auto_workflow_optimizer.py	Analyze and optimize workflow configurations	Weekly
Model Retraining	jobs/retrain_models.py	Retrain ML models with latest data	Daily at 2:00 AM
Webhook Processing	jobs/process_webhook_queue.py	Process queued webhook deliveries	Continuous
Health Monitoring	GET /api/health	Redis, MongoDB, MariaDB health probes	Every 15s (K8s)
Request Metrics	monitoring/metrics.py	Request counting and performance tracking	Continuous
Trace Generation	monitoring/tracing.py	UUID-based request tracing	Per request

3.2 Monitoring Infrastructure

Tool/Endpoint	Purpose	Access
/api/health	Service health (Redis, MongoDB, MariaDB, uptime, error rate)	Public (unauthenticated)
/api/metrics	Request/error counters	Authenticated

Tool/Endpoint	Purpose	Access
/api/system/status	Per-endpoint latency stats, detailed system health	Authenticated (admin)
/api/audit/log	Audit trail for security events	Authenticated (audit.view permission)
K8s Readiness/Liveness	/api/method/ping (15s interval, 5s timeout)	K8s internal
Docker Healthcheck	curl /api/health (30s interval)	Docker internal

3.3 Deployment Pipeline

Stage	Tool	Purpose
Code Quality	pre-commit, ruff, black	Lint and format enforcement
Unit Tests	pytest	Automated test execution
Security Scan	pip-audit, safety	Dependency vulnerability scanning
Build	Docker	Container image creation
Staging Deploy	Docker Compose / K8s	Pre-production validation
Production Deploy	K8s rolling update / GCP Cloud Run	Zero-downtime deployment
Post-Deploy Verify	/api/health + smoke tests	Production health validation

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4. Maintenance Schedules

4.1 Daily Maintenance (Automated)

Time (UTC)	Activity	Component

Time (UTC)	Activity	Component
02:00	ML model retraining	jobs/retrain_models.py
03:00	Full KPI metric refresh	jobs/nightly_kpi_refresh.py
03:30	Compliance audit scan	jobs/compliance_check.py
Continuous	Webhook queue processing	jobs/process_webhook_queue.py
Continuous	KPI streaming to WebSocket clients	server.py background scheduler
Every hour	KPI metric rollup/aggregation	jobs/kpi_rollup.py

4.2 Weekly Maintenance

Day	Activity	Owner
Monday	Review error rate trends from /api/metrics	Developer
Monday	Automated workflow optimization	jobs/auto_workflow_optimizer.py
Wednesday	Review AI alert false positive rates	Developer
Friday	Review webhook delivery success rates	Developer
Friday	Sprint retrospective and maintenance backlog grooming	Team

4.3 Monthly Maintenance

Activity	Owner	Duration
Dependency security scan and update	Developer	4 hours
Database index performance review	Developer	2 hours
MongoDB collection size audit and archival	DevOps	2 hours

Activity	Owner	Duration
Docker base image update	DevOps	2 hours
Configuration drift check	DevOps	1 hour
Backup restore verification	DevOps	2 hours
Code quality metrics review (complexity, duplication)	Developer	2 hours

4.4 Quarterly Maintenance

Activity	Owner	Duration
Frappe framework compatibility upgrade	Developer	8 hours
Database engine patch updates (MongoDB, MariaDB, Redis)	DevOps	4 hours
Load/stress testing on staging	Developer + DevOps	8 hours
Full security audit (OWASP Top 10 review)	Developer	8 hours
Dead code and unused dependency cleanup	Developer	4 hours
Disaster recovery drill (backup restore + failover)	Team	4 hours
RMMM plan review and risk re-assessment	Team	2 hours

5. Change Management Process

5.1 Change Categories

Category	Examples	Approval Required	Testing Required

Category	Examples	Approval Required	Testing Required
Standard	Config changes, minor bug fixes, dependency patches	Developer self-approval	Unit tests pass
Normal	New features, API changes, schema changes	Code review + Scrum Master	Full test suite + staging
Emergency	Security patches, critical bug fixes, incident response	Fast-track: Developer + Product Owner verbal	Targeted tests + post-deploy monitoring

5.2 Change Request Workflow

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Change Request -> Impact Assessment -> Approval -> Implementation -> Testing -> Staging -> Production -> Post-Im
  plementation Review
```

5.3 Rollback Strategy

Deployment Type	Rollback Method	RTO
Docker Compose	docker compose down && docker compose up with previous image tag	5 minutes
Kubernetes	kubectl rollout undo deployment/ibms-core	2 minutes
GCP Cloud Run	Revert to previous revision via Cloud Console	3 minutes
Database Schema	Run reverse migration script (Alembic downgrade / mongomigrate rollback)	15 minutes

6. Component Maintenance Matrix

6.1 Backend Components

Component	File/Module	Maintenance Focus	Update Frequency
FastAPI Server	server.py	Endpoint additions, middleware updates, security patches	Per sprint
Auth Engine	security/auth_engine.py	Password policy updates, rate limit tuning, RBAC changes	Monthly
JWT Auth	security/jwt_auth.py	Secret rotation, algorithm updates, expiry policy	Quarterly
Zero Trust	security/zero_trust.py	Service identity rules, network policy updates	Quarterly
AI Assistant	services/ai_assistant.py	NLP improvements, KPI context expansion	Per sprint
Anomaly Detection	services/anomaly.py	Model retraining, threshold calibration	Monthly (auto)
Fraud Detection	services/fraud_detection.py	Scoring model updates, threshold adjustments	Monthly
Decision Engine	services/decision_engine.py	Rule additions, threshold tuning	Per sprint
KPI Engine	services/kpi_engine.py	New metric definitions, computation logic	Per sprint
Dynamic Pricing	services/dynamic_pricing.py	Pricing formula updates, market factor weights	Monthly
Lead Scoring	services/lead_scoring.py	Scoring weight adjustments, new signal types	Monthly
Predictive Inventory	services/predictive_inventory.py	Demand model updates, safety stock calibration	Monthly
Risk Scoring Engine	risk_scoring_engine.py	Risk factor weights, new risk indicators	Monthly
Compliance Engine	compliance_engine.py	Regulatory rule updates, new compliance frameworks	Quarterly
Budget Optimizer	auto_budget_optimizer.py	Optimization algorithm refinement	Quarterly

Component	File/Module	Maintenance Focus	Update Frequency
Digital Twin	digital_twin.py	Simulation model updates, new scenario types	Quarterly

6.2 Database Components

Component	Technology	Maintenance Focus	Schedule
MongoDB	Motor + PyMongo	Index optimization, TTL policies, collection sharding	Monthly
MariaDB	SQLAlchemy + aiomysql	Schema migrations, query optimization, backup testing	Monthly
Redis	redis-py	Memory policy tuning, eviction strategy, persistence config	Quarterly

6.3 Frontend Components

Component	File	Maintenance Focus	Schedule
SPA Dashboard	frontend/index.html	Feature additions, accessibility, responsiveness	Per sprint
JavaScript App	frontend/static/js/app.js	New dashboard tabs, WebSocket enhancements, bug fixes	Per sprint
CSS/Tailwind	frontend/static/css/	Theme updates, design system maintenance	Per sprint

6.4 Infrastructure Components

Component	Configuration	Maintenance Focus	Schedule
Docker	Dockerfile, docker-compose.yml	Base image updates, security patches, optimization	Monthly

Component	Configuration	Maintenance Focus	Schedule
Docker Prod	docker-compose.prod.yml	Secret management, resource limits, SSL certs	Monthly
Kubernetes	deploy/k8s/	Manifest updates, resource right-sizing, HPA tuning	Quarterly
Nginx	nginx.conf	TLS config, rate limiting, header policies	Quarterly
AWS Terraform	deploy/aws/terraform/	Provider updates, resource changes, cost optimization	Quarterly
GCP Terraform	deploy/gcp/terraform/	Provider updates, service configuration	Quarterly

7. Knowledge Management

7.1 Documentation Maintenance

Document	Purpose	Update Trigger
README.md	Project overview, quickstart	Major feature additions
ARCHITECTURE.md	System architecture diagram	Architectural changes
API_DOCUMENTATION.md	API reference	Endpoint additions/changes
SECURITY_OVERVIEW.md	Security posture	Security control changes
DEPLOYMENT_GUIDE.md	Deployment procedures	Infrastructure changes
docs/system_blueprint.md	System design blueprint	Design changes
Sprint review docs (docs/sprint-review/)	Sprint deliverables and decisions	Every sprint

7.2 Runbook Requirements

Runbook Topic	Status	Priority
Incident response (SEV-1/SEV-2)	Planned	Critical
Database backup and restore	Planned	Critical
JWT secret rotation procedure	Planned	High
Kubernetes scaling and failover	Planned	High
MongoDB collection maintenance (TTL, archival)	Planned	Medium
New developer onboarding	Planned	Medium

8. Maintenance Metrics & KPIs

Metric	Target	Measurement
Mean Time to Repair (MTTR) -- SEV-1	< 4 hours	Incident tracker
Mean Time to Repair (MTTR) -- SEV-2	< 24 hours	Incident tracker
System Uptime	>= 99.5%	/api/health monitoring
Deployment Success Rate	>= 95%	CI/CD pipeline metrics
Test Pass Rate	100% (critical/high)	pytest CI reports
Dependency CVE Exposure	0 critical, < 5 high	pip-audit monthly scan
Change Failure Rate	< 10%	Post-deploy incident correlation
Mean Time Between Failures (MTBF)	> 720 hours	Incident frequency analysis
Code Quality Score	A (SonarQube or similar)	Monthly scan

Metric	Target	Measurement
Documentation Currency	Updated within 1 sprint of change	Manual review

9. Maintenance Roles & Responsibilities

Role	Responsibilities
Product Developer	Corrective fixes, perfective enhancements, code quality, test maintenance, dependency updates, model retraining oversight
Product Owner	Change prioritization, feature roadmap alignment, user feedback triage, maintenance budget allocation
Scrum Master	Process enforcement, sprint planning for maintenance items, escalation management, team capacity planning
DevOps (future)	Infrastructure maintenance, deployment automation, monitoring, backup management, security patching

10. End-of-Life (EOL) Planning

Consideration	Plan
Data retention policy	KPI Snapshots: 2 years active + 5 years archived. Audit logs: 7 years. Webhook logs: 90 days.
Data export	Provide API and CLI tools for bulk data export in JSON/CSV format

Consideration	Plan
Migration path	Document data schema and API contracts for successor system integration
User notification	6-month advance notice for any module deprecation
Dependency EOL tracking	Monitor Python, FastAPI, Frappe, MongoDB, MariaDB, Redis EOL timelines

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11. Approval

Role	Name	Date
Product Developer	Shivansh Srivastava	April 14, 2026
Product Owner	Prahallad Padhan	April 14, 2026
Scrum Master	Ranveer Rai Khare	April 14, 2026